Prashant Rai

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Summary

AI/ML Engineer currently pursuing B.Tech in Computer Science (AI), experienced in building GenAI/LLM-based solutions using Prompt Engineering, Agentic AI, and LLM APIs (OpenAI, Groq, Claude, Gemini). Skilled in Python (FastAPI/Flask), machine learning, time-series forecasting, and end-to-end AI product development with React/Next.js frontends. AWS Certified AI Practitioner with hands-on experience benchmarking model performance and documenting findings.

Education

B.Tech in Computer Science and Engineering (Artificial Intelligence) Expected 2027
KIET Group of Institutions, AKTU CGPA: **7.5**

Senior Secondary Education (CBSE) 2022
Divine Sainik School **78%**

Secondary Education (CBSE) 2020
Glenhill School **85%**

Experience

Cloud & Data Engineering Intern (AWS Capstone) Dec 2025 – Feb 2026
Python, Flask, AWS EC2, DynamoDB, SNS GitHub

- Built and deployed a full-stack cloud platform for agricultural crop yield data (10,000+ records) using Python, Flask, and AWS (EC2, DynamoDB, SNS, IAM).
- Optimized DynamoDB queries via a Global Secondary Index, cutting latency by 60.5% (4.19s → 1.65s) vs. table scans.
- Built a CI/CD pipeline (GitHub Actions → EC2), cutting deployment time from ∼20–25 min (manual) to ∼13 sec.
- Designed an S3 + Athena data lake, cutting analytics query time by ∼83% (5.9x) vs. DynamoDB scans.
- Trained a RandomForest yield-forecasting model (19,689 records, R^2=0.9735) and built an agentic Gemini LLM assistant for natural-language queries across the platform.

AI & Data Science Intern (XYlofy AI) Jun 2026 – Jul 2026
Python, Machine Learning, LLM APIs, Time Series Analysis, Scikit-Learn, Streamlit GitHub

- Developed an end-to-end Sales Forecasting System using SARIMA, Prophet, and XGBoost models, benchmarking model performance to predict multi-category demand trends.
- Built an Employee Attrition Prediction model (Classification algorithms) and implemented Anomaly Detection (Isolation Forest) plus K-Means Clustering for product demand segmentation.
- Deployed an interactive Streamlit web dashboard (Python/React-style component integration) to showcase real-time business metrics and predictive analytics.

Projects

Hybrid Renewable Energy Optimization System Jun 2026 – Present
Python, Machine Learning, IoT, Energy Optimization, Data Analytics GitHub

- Designing a hybrid energy system integrating solar, wind, and battery storage for efficient and reliable power generation.
- Developing machine learning models to forecast energy output using environmental inputs like solar irradiance and wind speed.
- Implementing real-time optimization algorithms to dynamically switch energy sources and simulating smart grid demand-supply balance to minimize power loss and improve efficiency.
- Simulating smart grid behavior to balance energy demand and supply, improving system efficiency and sustainability.

AgenticPay – Autonomous AI Agent Payment System Jun 2026 – Aug 2026
Python, Flask, Next.js, Prompt Engineering, LLM APIs (Groq/Llama 3), Web3.py, Avalanche, x402, ERC-8004, Tailwind GitHub

- Built an autonomous multi-agent payment system where AI agents hire, verify, and pay each other on Avalanche blockchain using x402 HTTP payment protocol – zero human intervention required.
- Implemented ERC-8004 on-chain agent identity and reputation registry; low reputation agents are automatically blocked from receiving payments, preventing impersonation.
- Deployed 23+ real AVAX transactions on Avalanche Fuji Testnet,with Groq AI (Llama 3) powering real-time research and report generation – fully verifiable on Snowtrace.

Skills

GenAI / LLM: Prompt Engineering, LLM APIs (OpenAI, Claude, Groq, Gemini), LLM Agents, Model Benchmarking
Languages: Python, SQL, JavaScript, C, Java
AI / Machine Learning: Machine Learning, Deep Learning basics, Time Series Forecasting (SARIMA, Prophet, XGBoost), Classification, Clustering, Anomaly Detection, Feature Engineering, Scikit-Learn
Backend / API Development: FastAPI, Flask, Node.js, REST APIs, Streamlit, Firebase
Frontend / Integration: React.js, Next.js, Tailwind, Bootstrap, HTML/CSS
Data & Analytics: Data Analysis, Predictive Analytics, PowerBI, Pandas, NumPy, API Data Processing, Structured & Unstructured Data
Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, H2
Cloud & Tools: AWS (EC2, DynamoDB, SNS), Git, GitHub, Web3.py, AWS Certified

Certifications

- Skills to Jobs: AWS Cloud & Generative AI Capstone – Smartbridge x AWS *[Certificate]* Apr 2026
- AWS Certified AI Practitioner – Amazon Web Services *[Certificate]* Nov 2025
- AWS Academy Graduate – Generative AI Foundations *[Certificate]* Sep 2025